

Degree: B.Tech, Stream: EEE

Sixth Semester Assignment,

Faculty Name: Prof. Dr. Manas Mukherjee

Subject Name: Power System II

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Full Marks: 50

13*2=26

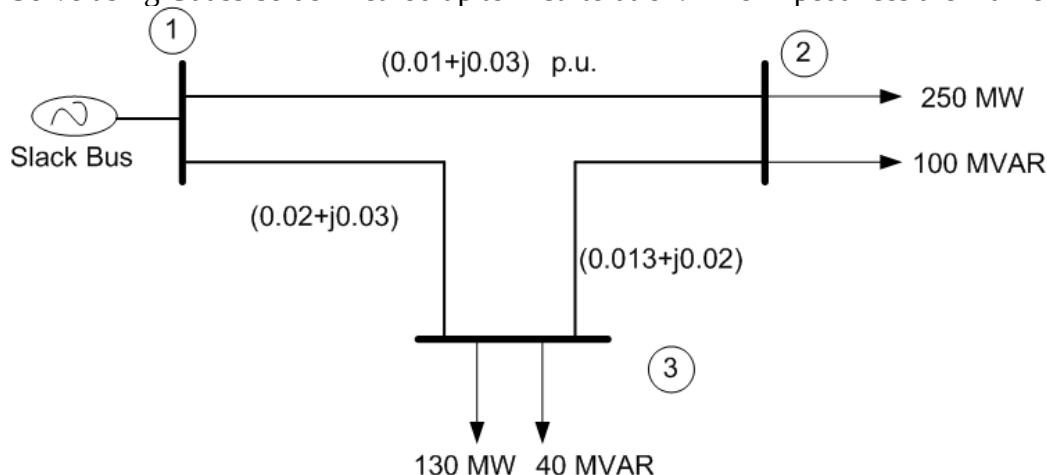
Group A

1. What is slack bus? Why is it required?
2. What are the properties of Bus Admittance matrix?
3. What is acceleration factor? What is its significant?
4. Derive the expression of injected active and reactive power (P_i, Q_i) for a bus.
5. Why Y bus matrix is preferred in load flow analysis instead of Z bus matrix. Write advantages of Y bus matrix.
6. In 50 bus power system, find the size of Jacobian Matrix with the following information: -
Slack bus- 1; Generator Bus- 8; Load Bus- 41;
7. Describe the factors, responsible for influencing the convergence rate in GS load flow method.
8. Why load flow is necessary?
9. Compare different low flow methods.
10. For a 25-bus system with 5 generator buses and 20 load buses, find the size of Jacobian matrix.
11. Write the types of buses with their known and unknown quantities.
12. Describe slack or swing bus.
13. Discuss the advantages of sparse matrix in load flow analysis.

2*12=24

Group B

1. Solve using Gauss-seidel method up to first iteration. Line impedances are marked on 100 MVA base.



2. Write Flow chart of Gauss – Siedel method for load flow analysis of a network with PV buses.